SUSTAINABILITY
Complete Domain Learning Guide

10 modules • Clear explanations • Worked examples • Checkpoints • Practice assessment

A practical guide to sustainability concepts, emissions accounting, life-cycle thinking, circular systems, resource efficiency, ESG reporting, intervention design and careers.

Practice assessment: 20 questions
Suggested pass target: 14/20 (70%)

How to use this guide

Read each module in order. Focus on the distinctions between similar terms, because scenario-based questions test how concepts are applied. Work through the calculations, answer each checkpoint, and finish with the 20-question practice assessment.

Core principle: Sustainability claims should be tied to a defined boundary, baseline, credible data, time period and transparent method. Good intentions are not the same as measured results.

1. Sustainability Fundamentals

Sustainability across three dimensions

Sustainability means meeting present needs without undermining future generations' ability to meet their own. Environmental sustainability protects climate, ecosystems and resources; social sustainability supports health, safety, equity, rights and inclusion; economic sustainability creates durable livelihoods and value without exhausting the natural and social systems on which business depends. A project is not fully sustainable just because it is profitable or low-carbon if it harms workers or destroys habitats.

Planetary boundaries

Planetary boundaries are scientifically proposed limits for major Earth-system processes. They include climate change, biosphere integrity, land-system change, freshwater change, nitrogen and phosphorus flows, ocean acidification, stratospheric ozone depletion, atmospheric aerosols and novel entities such as persistent synthetic chemicals. Crossing a boundary raises the risk of large-scale or irreversible change; these are risk guides, not simple local thresholds.

Sustainable Development Goals (SDGs)

The UN adopted 17 SDGs in 2015 as part of the 2030 Agenda. They cover poverty, hunger, health, education, gender equality, water, clean energy, decent work, industry, inequality, cities, responsible consumption, climate, oceans, land ecosystems, peace and partnerships. Organisations should connect projects to real outcomes rather than merely display SDG icons.

Triple Bottom Line and CSR

The Triple Bottom Line (TBL) considers people, planet and profit. Corporate social responsibility (CSR) concerns an organisation's responsibility for the effects of its decisions on society and the environment, often through policies, programmes and engagement. TBL is a way to assess performance across three dimensions; CSR is a broader responsibility and management approach. Sustainability should influence core operations, not only charity.

Scenario

A factory increases profit by using cheaper fuel, but emissions rise and workers report unsafe conditions. TBL examines financial performance, environmental effects and worker wellbeing together. CSR asks what responsibilities, safeguards, consultation and corrective actions the factory should implement.

Module checkpoint
Which framework asks an organisation to assess people, planet and profit together?

Answer: Triple Bottom Line. CSR also matters for organisational responsibilities and response.

2. Climate Systems & Emissions

Greenhouse effect

Sunlight reaches Earth; some energy is reflected and the rest warms the surface. Earth emits infrared radiation. Greenhouse gases such as carbon dioxide (CO■), methane (CH■), nitrous oxide (N■O) and certain industrial gases absorb and re-emit some infrared radiation, warming the lower atmosphere and surface. The natural greenhouse effect makes Earth habitable; the concern is the enhanced effect caused by human activities.

Radiative forcing

Radiative forcing describes a change in Earth's energy balance, often expressed in watts per square metre. Positive forcing tends to warm the climate; negative forcing tends to cool it. More greenhouse gases generally create positive forcing. Emissions are flows released over time, while atmospheric concentration is the amount present; they are related but not interchangeable.

Scope 1, 2 and 3

Scope 1: direct emissions from owned or controlled sources, such as fuel burned in a company boiler or company vehicle. Scope 2: indirect emissions associated with purchased electricity, steam, heating or cooling. Scope 3: other upstream and downstream value-chain emissions, such as purchased materials, business travel, commuting, product use and end-of-life. Scope 3 can be the largest category for many organisations.

Reading an inventory

An emissions inventory lists sources, activity data, factors, calculated emissions, scopes, reporting period, boundaries and methods. Check whether the same facilities and periods are compared, whether estimates are disclosed, and whether changes reflect real reductions or changed business size, methodology or boundaries.

Scenario

Natural gas burned in an office boiler is Scope 1; purchased grid electricity is Scope 2; employee flights and purchased laptops are usually Scope 3. Classification depends on source and organisational boundary, not simply where emissions physically occur.

> **Module checkpoint**
> Classify company-controlled van fuel, purchased electricity and employee air travel.

Answer: Scope 1; Scope 2; Scope 3, respectively.

3. Carbon Footprinting

Activity data and emission factors

Activity data quantify what happened: litres of diesel, kWh of electricity, kilograms of material or kilometres travelled. An emission factor converts activity into estimated emissions, such as kg CO■e per kWh. Match the factor to activity, geography, technology, unit and reporting period.

Core calculation

Emissions = activity data × emission factor. Convert units carefully, sum comparable sources and document assumptions. If the result is in kilograms, divide by 1,000 to convert to tonnes.

Carbon dioxide equivalent (CO■e)

CO■e expresses the warming effect of different greenhouse gases in a common unit using a selected global-warming-potential time horizon. Different gases have different warming effects per kilogram. State the factors and time horizon used because results can vary by method.

Organisational boundary

Define which legal entities, facilities, assets and operations are included. Common approaches include operational control, financial control or equity share. Acquisitions or disposals can make year-to-year comparisons misleading unless historical data are recalculated or the change is explained.

Uncertainty and data quality

Meter readings and invoices are generally stronger than rough estimates, but all data have limitations. Record source, date, unit, estimation method and uncertainty. Avoid false precision: a footprint is an evidence-based estimate, not an exact physical count.

Worked calculation

A site uses 12,000 kWh. If an illustrative factor is 0.70 kg CO■e/kWh, emissions = 12,000 × 0.70 = 8,400 kg CO■e = 8.4 t CO■e. For real inventories, use an appropriate published factor.

> **Module checkpoint**
> A site uses 5,000 litres of fuel at an illustrative factor of 2.5 kg CO■e/litre. Calculate emissions.

Answer: 5,000 × 2.5 = 12,500 kg CO■e = 12.5 t CO■e.

4. Life Cycle Assessment (LCA)

Goal and scope

LCA evaluates environmental impacts of a product or service across defined life stages. The goal explains why the study is being performed and for whom. Scope defines the product system, boundaries, impact categories, data needs and assumptions.

Functional unit

The functional unit is the quantified service being compared. Comparing detergent bottles alone may be unfair if one bottle washes twice as many loads. A fairer unit could be ‘washing 100 standard loads to a specified cleaning performance’.

Cradle to grave

A cradle-to-grave study includes raw material extraction, processing, manufacturing, transport, use and end-of-life. Cradle-to-gate ends when a product leaves the factory; gate-to-gate studies one stage. Boundaries should fit the decision and exclusions must be disclosed.

Inventory and impact assessment

The life-cycle inventory tracks inputs such as energy, water and materials and outputs such as emissions and waste. Impact assessment translates these flows into potential impacts such as climate change, water use, acidification or eutrophication. One impact category rarely tells the full story.

Burden shifting

Burden shifting occurs when an improvement in one stage increases impact elsewhere. An electric product may have no exhaust emissions during use but still has manufacturing, electricity-generation and end-of-life impacts. Reusable products help only when use counts and cleaning or transport practices support the benefit.

Scenario

To compare paper and reusable cups, use the same functional unit and include production, washing, transport and disposal. Results depend on energy mix, washing practices and actual reuse count.

Module checkpoint
What makes a product comparison fair in an LCA?

Answer: Use a common functional unit representing the same service and performance.

5. Circular Economy

Linear versus circular

A linear economy follows take–make–use–dispose. A circular economy designs waste and pollution out, keeps products and materials in use at high value, and supports natural-system regeneration. Recycling matters, but preventing waste and extending product life can retain more value.

Hierarchy of actions

A practical priority order is refuse or rethink unnecessary demand, reduce material and energy use, reuse, repair, refurbish or remanufacture, and recycle when higher-value options are not practical. Recycling should not justify avoidable consumption.

Design for disassembly

Products designed for disassembly use accessible fasteners, labelled materials, modular components and repairable parts. Avoid permanent bonding of incompatible materials where possible. Consider durability, spare parts, safety and recovery from the beginning.

Product as a service

Customers pay for use or access rather than ownership, for example lighting as a service or leased equipment. Retained ownership may incentivise durability and efficiency, but transport, maintenance, increased demand and end-of-life still matter.

Scenario

A company can extend laptop life through replaceable batteries, repairs, secure data wiping, refurbishment and resale. Recycling is appropriate for devices that cannot safely be reused.

> **Module checkpoint**
> A phone has a replaceable battery. What should be considered before recycling?

Answer: Repair or replace the battery and continue using the phone if safe and practical.

6. Energy & Resource Efficiency

Energy audit

An energy audit systematically identifies where energy is used, when demand occurs and which changes could reduce consumption. Gather bills and meter data, inspect equipment and schedules, identify significant loads, estimate savings, rank measures, then verify savings after implementation.

Baseline consumption

A baseline is the measured starting point before an intervention. Account for weather, occupancy, production volume and operating hours. A building using less energy because it was closed is not necessarily more efficient.

Simple payback

Simple payback = upfront cost ÷ annual net savings. If LED upgrades cost ■2,40,000 and save ■80,000 per year, payback is 3 years. This ignores the time value of money, equipment life, maintenance, energy-price changes and other benefits.

Marginal abatement cost

Marginal abatement cost estimates cost per additional unit of emissions avoided. A simplified form is (annualised incremental cost – annual savings) ÷ annual emissions avoided. A negative cost can mean a measure saves money while reducing emissions. State assumptions and period.

Rebound effect

Efficiency improvements can lower the cost of a service and encourage more use, partly offsetting savings. A more efficient air conditioner may be run longer or at a lower temperature. Pair efficiency with sensible operation and track total consumption.

Renewables realistically

Assess solar or wind resources, site suitability, grid connection, storage, tariffs, maintenance and life-cycle impacts. Renewable procurement claims should match the contract and accounting method. Reduce avoidable demand, then choose a suitable supply strategy.

> **Module checkpoint**
> A project costs ■1,50,000 and saves ■50,000 annually. What is simple payback?

Answer: ■1,50,000 ÷ ■50,000/year = 3 years.

7. Water, Waste & Biodiversity

Water footprint and stress

A water footprint measures freshwater consumed or polluted through a product or operation, including relevant supply-chain stages. Location matters: one litre in a water-stressed basin may have different consequences from one litre where water is abundant. Withdrawal is water taken from a source; consumption is water not returned to the same source in the same condition and time.

Waste diversion

Waste diversion is the proportion kept from landfill or other defined disposal routes through reuse, recycling or composting. A simple rate is diverted waste ÷ total waste generated × 100%. Weigh materials where possible and verify that separated waste reaches the intended processor.

Waste segregation

Separate organics, paper, metals, glass, plastics, e-waste, batteries and hazardous materials according to local rules and available systems. Contamination can make recyclables unusable. Batteries and electronic waste require appropriate handling.

Biodiversity

Biodiversity includes variation within species, between species and across ecosystems. Projects can affect it through habitat loss, fragmentation, pollution, invasive species, resource overuse and climate change. Avoiding impacts is generally preferable to restoring damage later.

Ecosystem services

Ecosystems provide provisioning services (food and freshwater), regulating services (pollination, flood control and carbon storage), cultural services (recreation and cultural meaning) and supporting ecological processes.

Scenario

Before a warehouse expansion, assess water stress, drainage, habitats, species, waste and community impacts. Consider already-developed land, efficient fixtures, rainwater management, habitat buffers and monitoring.

> **Module checkpoint**
> A site diverts 360 kg of 600 kg total waste. What is its diversion rate?

Answer: 360 ÷ 600 × 100% = 60%, assuming the metric is clearly defined.

8. ESG & Reporting Frameworks

ESG

Environmental, social and governance (ESG) is a lens for evaluating organisational risks, impacts and management. Environmental issues include emissions, water and biodiversity; social issues include worker safety, labour rights and communities; governance includes ethics, oversight and accountability. It is not one universal score.

Major frameworks

GRI Standards focus on an organisation’s impacts on the economy, environment and people. IFRS Sustainability Disclosure Standards, including IFRS S1 and S2, address sustainability-related financial information useful to investors, with S2 focused on climate. The GHG Protocol provides widely used greenhouse-gas accounting standards. Check current local rules and applicability.

Materiality assessment

Materiality identifies sustainability topics that matter for decision-making. Impact materiality asks how the organisation affects people and nature. Financial materiality asks how sustainability risks and opportunities affect enterprise value. Double materiality considers both perspectives. Identify stakeholders, gather evidence, score issues transparently and document decisions.

Assurance

Assurance is an independent examination of sustainability information. Limited assurance involves less extensive work than reasonable assurance; neither guarantees every number is perfect. Source records, controls and traceable calculations help.

Greenwashing

Greenwashing presents environmental performance as better than evidence supports. Warning signs include vague ‘eco-friendly’ language, irrelevant claims, unverified badges, cherry-picked data, missing boundaries and future goals without credible plans. Prefer specific, comparable, time-bound and verifiable claims.

Scenario

For a ‘carbon neutral’ product, ask which emissions are included, whether value-chain reductions occurred, what offsets were used, whether claims were checked and how remaining emissions are explained.

> **Module checkpoint**
> A company calls packaging ‘100% green’ without evidence. What is the concern?

Answer: Potential greenwashing: the claim is vague and unsupported.

9. Designing an Intervention

Theory of Change

A Theory of Change explains how activities are expected to lead to outputs, outcomes and longer-term impact, including assumptions and external factors. Example: install efficient pumps → lower electricity use → lower emissions and operating costs, provided service levels remain comparable.

Baseline, target and timeline

Measure the starting condition before action. A strong target specifies metric, baseline, target value, deadline, organisational boundary and method. ‘Reduce electricity use by 15% by December 2027 against a 2026 baseline, adjusted for production volume’ is stronger than ‘be greener’.

Stakeholders and incentives

Identify people affected by or able to influence the intervention: workers, customers, suppliers, communities, regulators, managers and funders. Understand their concerns and incentives. Co-design can reveal barriers and improve trust.

Monitoring and verification

Monitoring tracks progress at planned intervals; verification checks whether reported results are supported by evidence and methods. Define data owners, frequency, quality checks, documentation, comparison method and corrective actions before starting.

Co-benefits and trade-offs

A co-benefit is an additional positive result, such as energy efficiency reducing costs and improving comfort. Trade-offs can include upfront cost, land use, supply-chain impacts or unequal benefits. Make these explicit and plan safeguards.

Worked intervention plan

Goal: cut campus food waste. Baseline: weigh waste daily for four weeks and record meals served. Intervention: improve demand forecasting, offer portion choice and donate safe surplus under applicable rules. Target: reduce avoidable waste per meal by 25% in six months. Monitor waste weights, meal counts, food safety and donation records monthly.

Module checkpoint
Why establish a baseline before starting an intervention?

Answer: It provides the starting condition needed to assess change.

10. Careers in Sustainability

Sustainability analyst

Collects and validates environmental and social data, maintains indicators, analyses trends, prepares reports and supports improvement plans. Useful skills include spreadsheets, visualisation, documentation, basic statistics, emissions accounting and communication.

Carbon accounting

Defines organisational boundaries, collects activity data, selects emission factors, calculates Scope 1/2/3 emissions, documents assumptions and supports reporting or assurance. Units, evidence and consistency are essential.

Impact measurement

Evaluates what changed because of a programme, for whom, by how much and whether the programme contributed. It distinguishes activities and outputs from outcomes and impacts and considers unintended effects. A participant count alone does not prove long-term impact.

Stakeholder engagement

Identifies affected groups, listens to their needs, shares information accessibly, responds to concerns and documents how input shaped decisions. Meaningful engagement requires inclusive methods and feedback.

Portfolio projects

Create an illustrative emissions inventory, energy audit with payback calculations, waste-diversion dashboard, life-cycle comparison with assumptions, or sustainability report mock-up. Clearly separate real measurements from simulated data.

Career readiness

Practise spreadsheet formulas, data cleaning, charts, concise writing, source tracking, stakeholder mapping and explaining limitations. Strong interview answers cover problem, method, assumptions, result, limitations and next step.

Module checkpoint
Which role commonly applies emission factors and calculates organisational GHG emissions?

Answer: A carbon accounting specialist or sustainability professional responsible for GHG inventories.

Cross-module worked examples

1. Calculate and classify emissions

A business reports natural gas burned in its boiler = 4,000 kg CO■e; purchased electricity = 7,500 kg CO■e; employee flights = 2,000 kg CO■e. Total = 13,500 kg CO■e (13.5 t CO■e). Boiler fuel is Scope 1, electricity Scope 2, and flights Scope 3. This total is valid only if reporting period, boundary and methods are consistent.

2. Prioritise an efficiency measure

Measure A costs ■1,20,000 and saves ■40,000 per year: 3-year payback. Measure B costs ■90,000 and saves ■15,000 per year: 6-year payback. A has shorter payback, but also compare emissions avoided, equipment life, maintenance, comfort, safety and uncertainty.

3. Set a measurable intervention

Weak target: ‘Use less water.’ Stronger target: ‘Reduce potable water use per occupied person by 12% by the end of 2027 compared with the 2026 baseline, using monthly meter readings and documenting occupancy.’ This specifies metric, comparison, baseline, deadline and evidence.

Practice assessment — 20 questions

Choose one best answer per question. The answer key includes explanations. A score of 14/20 equals 70%.

1. Which statement best defines sustainability?

A. Maximise current output regardless of future costs

B. Meet present needs while protecting future generations’ ability to meet theirs

C. Focus only on recycling

D. Replace all business activity with conservation

2. Which framework considers people, planet and profit?

A. Triple Bottom Line

B. Scope 2

C. Functional unit

D. Assurance

3. Fuel burned in a company-controlled boiler is usually:

A. Scope 1

B. Scope 2

C. Scope 3

D. Outside every inventory

4. Emissions associated with purchased electricity are generally:

A. Scope 1

B. Scope 2

C. Scope 3 only

D. Not measurable

5. Employee business flights are generally:

A. Scope 1

B. Scope 2

C. Scope 3

D. Planetary boundaries

6. Which formula estimates emissions?

A. Activity ÷ factor

B. Activity × emission factor

C. Factor – activity

D. Activity + reporting year

7. What does CO■e do?

A. Measures only carbon monoxide

B. Expresses different greenhouse gases in a common warming-equivalent unit

C. Measures water use

D. Describes waste diversion

8. What is an organisational boundary?

A. A building fence

B. The operations and facilities included in an inventory

C. The target year

D. A type of factor

9. A functional unit in LCA is:

A. A marketing slogan

B. A quantified service used to compare alternatives fairly

C. Annual profit

D. A waste-bin label

10. A cradle-to-grave assessment includes:

A. Manufacturing only

B. Raw materials through use and end-of-life

C. Disposal only

D. Office electricity only

11. Burden shifting occurs when:

A. All impacts decline

B. An improvement in one stage increases impact elsewhere

C. Emissions are measured

D. Waste is weighed

12. Which usually preserves product value better than recycling?

A. Repairing a product so it can be used longer

B. Sending it to landfill

C. Mixing waste streams

D. Replacing it immediately

13. A baseline is:

A. The measured starting condition before an intervention

B. An aspiration with no metric

C. An assurance provider

D. A product label

14. A project costs ■80,000 and saves ■20,000 yearly. Simple payback is:

A. 2 years

B. 3 years

C. 4 years

D. 5 years

15. The rebound effect is:

A. Efficiency gains partly offset by increased use

B. An assurance audit

C. A biodiversity survey

D. A Scope 1 source

16. Why does local water stress matter?

A. Water impacts are identical everywhere

B. The same volume can have different consequences in different basins

C. Water footprint measures only rainfall

D. It removes the need to measure

17. Which is an ecosystem service?

A. Pollination by insects

B. A corporate org chart

C. Software version number

D. An emission factor

18. Which is a greenwashing warning sign?

A. A defined, checked metric

B. A clear boundary and period

C. A vague ‘eco-friendly’ claim without evidence

D. Publishing limitations

19. Monitoring and verification differ because:

A. Monitoring tracks progress; verification checks evidence and claims

B. Both mean advertising

C. Verification replaces data collection

D. Monitoring only happens at the end

20. Which is the strongest target?

A. Become sustainable soon

B. Improve water

C. Reduce water use by 12% per occupied person by end-2027 against a 2026 baseline, measured monthly

D. Tell staff to care more

Answer key with explanations

1. B. It includes present and future needs.

2. A. TBL assesses social, environmental and economic performance.

3. A. It is a direct emission source controlled by the organisation.

4. B. Scope 2 covers purchased electricity, steam, heating and cooling.

5. C. They are value-chain emissions.

6. B. The factor converts activity into emissions.

7. B. It allows greenhouse gases to be aggregated using equivalence factors.

8. B. Boundary rules determine which operations are included.

9. B. It aligns comparisons around equivalent function.

10. B. It covers the full life cycle.

11. B. An improvement should be assessed across relevant stages and impacts.

12. A. Repair retains the product and embedded manufacturing value.

13. A. It supports before-and-after comparison.

14. C. ■80,000 ÷ ■20,000/year = 4 years.

15. A. Lower operating costs can encourage more consumption.

16. B. Location and scarcity influence water impacts.

17. A. Pollination is a regulating ecosystem service.

18. C. Vague unsupported claims can mislead.

19. A. Monitoring measures; verification checks support for reported results.

20. C. It includes a metric, baseline, deadline and method.

Quick revision sheet

Term	Remember
Sustainability	Balance environmental, social and economic needs across generations.
Planetary boundaries	Risk limits for major Earth-system processes.
SDGs	17 UN goals for sustainable development by 2030.
Triple Bottom Line	People, planet, profit.
Scope 1 / 2 / 3	Direct controlled sources / purchased energy / other value-chain emissions.
Emissions calculation	Activity data × emission factor; kg to tonnes: divide by 1,000.
CO■e	Common unit for the warming effect of different greenhouse gases.
LCA	Assess life-cycle impacts; define goal, scope and functional unit.
Burden shifting	A reduction in one stage can increase impact elsewhere.
Circular economy	Keep products/materials in use; prevent, reuse and repair before recycling where feasible.
Energy audit	Identify where energy is used and which changes are worthwhile.
Simple payback	Upfront cost ÷ annual net savings.
Water footprint	Freshwater consumed or polluted through a product or operation.
ESG	Environmental, social and governance performance and risk lens.
Materiality	Identify issues important to impacts and/or enterprise value.
Greenwashing	Claims exceed the evidence or omit important context.
Theory of Change	Explain how activities are expected to lead to outcomes and impact.
Baseline and target	Starting measurement + quantified goal + deadline + method.
Co-benefit	Additional positive outcome beyond the main objective.
Impact measurement	Assess actual change, not just activities or participation counts.

Exam tip: First identify whether a scenario asks about a concept, emissions scope, calculation, life-cycle boundary, efficiency, disclosure quality or intervention design. State assumptions, show units and prefer measurable evidence over broad claims.